

Excel Mastery Challenge: Sales Data Analysis

Assignment Overview

Welcome to your Excel Mastery Challenge! This assignment is designed to test and enhance your skills across a wide range of Excel functionalities, from data cleaning and manipulation to advanced analysis and visualization. You will be working with a simulated sales dataset for a retail company, tasked with extracting key insights to inform business decisions. This is not a simple task; it requires careful thought, application of various functions, and a keen eye for detail.

Objective: Analyze a raw sales dataset to identify trends, top performers, and areas for improvement, presenting your findings in a clear and actionable manner.

Deliverables:

1. A cleaned and processed Excel workbook (`Sales_Analysis_YourName.xlsx`).
2. A summary sheet with key metrics and insights.
3. A dashboard sheet with interactive charts and PivotTables.

Part 1: Data Preparation and Cleaning (Module 1 & Optional)

Your raw data will likely contain inconsistencies and require preparation before analysis.

1. **Import & Inspect:** Assume you have imported a CSV file. Inspect the data for common issues (e.g., inconsistent formatting, missing values, duplicates).
2. **Data Type Conversion:** Ensure all columns are correctly formatted (e.g., `Order Date` as Date, `Sales Amount` as Currency, `Quantity` as Integer).
3. **Remove Duplicates:** Identify and remove any duplicate transaction records based on a combination of `Order ID` and `Product ID` .
4. **Text Cleaning:** Standardize `Product Category` and `Region` names (e.g., ensure consistent casing, remove leading/trailing spaces using `TRIM()`).
5. **Add Helper Columns:**
 - Create a `Year` column using the `YEAR()` function from `Order Date` .
 - Create a `Month Name` column using a combination of `TEXT()` and `MONTH()` functions.

- Create a `Day of Week` column using `TEXT()` or `WEEKDAY()` .

Part 2: Core Data Analysis (Module 2 & 3)

Apply various functions to extract meaningful information from your cleaned dataset.

1. Basic Statistics:

- Calculate the `Total Sales Amount` using `SUM()` .
- Determine the `Average Order Value` using `AVERAGE()` .
- Find the `Number of Unique Products Sold` using `COUNTA()` on a unique list of products.
- Count the `Number of Orders` using `COUNT()` or `COUNTA()` .
- Identify the `Median Quantity Sold` per order using `MEDIAN()` .

2. Cell Referencing Practice:

- In a separate sheet, create a small table for `Discount Rate` (e.g., 5%, 10%, 15%). Use **absolute references** when applying these rates to calculate `Discount Amount` for various scenarios.

3. Logical Functions:

- Create a new column `Order Status` . Use an `IF()` function to categorize orders: if `Sales Amount` is greater than \$500, it's "Large Order"; otherwise, it's "Standard Order".
- Use `AND()` and `OR()` functions to identify specific types of orders, e.g., "High Value, High Quantity" orders (`Sales Amount` > \$750 AND `Quantity` > 5) or "Weekend Orders" (`Day of Week` is Saturday OR Sunday).

4. Conditional Formatting:

- Apply conditional formatting to the `Sales Amount` column to highlight values above the average in green and below average in red.
- Use data bars on the `Quantity` column to visually represent order sizes.
- Highlight duplicate `Customer IDs` (if any) to identify potential data entry errors.

5. Data Validation:

- On a new input sheet, create a cell for `Region` selection. Use Data Validation to create a dropdown list of unique regions from your dataset.
- For a `Target Sales` input cell, use Data Validation to ensure only whole numbers between 100 and 10000 are entered.

6. Lookup Functions:

- Assume you have a separate **Product_Lookup** table with **Product ID** and **Product Cost** . Use **XLOOKUP()** (or **VLOOKUP()** if you prefer) to retrieve the **Product Cost** for each transaction in your main sales data.
- Calculate **Profit** for each transaction (**Sales Amount** - **Product Cost**).

7. Criteria-Based Statistical Functions:

- Calculate **Total Sales for 'Electronics' Category** using **SUMIF()** .
- Calculate **Average Sales for 'North' Region** using **AVERAGEIF()** .
- Count **Orders with Quantity > 5** using **COUNTIF()** .
- Calculate **Total Profit for 'Clothing' Category in 'West' Region** using **SUMIFS()** .

Part 3: Advanced Analysis & Visualization (Module 3 & 4)

Leverage Excel's powerful analytical and visualization tools.

1. PivotTables:

- Create a PivotTable to summarize **Total Sales** by **Product Category** and **Region** .
- Add **Year** as a filter to analyze sales trends over time.
- Create another PivotTable to show **Average Quantity Sold** by **Customer Segment** .
- Use Slicers to make your PivotTables interactive.

2. Charts & Data Visualization:

- Based on your PivotTable data, create a **Column Chart** showing **Total Sales by Product Category** .
- Create a **Line Chart** to visualize **Monthly Sales Trends** (using your **Month Name** and **Year** columns).
- Create a **Bar Chart** showing **Top 5 Products by Sales Amount** .
- Ensure all charts are clearly labeled with titles and axis labels.

Part 4: Efficiency & Protection (Optional, but Recommended)

Demonstrate your understanding of best practices.

1. **Tables (Ctrl + T):** Convert your main data range into an Excel Table. Explain the benefits.
2. **Freeze Panes:** Apply Freeze Panes to keep your headers visible while scrolling through the data.

3. **Protect Sheet:** Protect your summary and dashboard sheets to prevent accidental changes to formulas or charts, allowing users to interact only with filters and slicers.
4. **Basic Formatting Shortcuts:** Briefly describe how you used any formatting shortcuts during your assignment.

Submission Guidelines

- Save your workbook as `Sales_Analysis_YourName.xlsx` .
- Ensure all calculations are dynamic (i.e., use formulas, not hard-coded values).
- Organize your workbook with clearly named sheets (e.g., `Raw Data` , `Cleaned Data` , `Summary` , `Dashboard` , `Input`).
- Provide brief explanations for your choices in conditional formatting, data validation, and chart types.

Good luck, and have fun mastering Excel!